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Module 6 Assignment: Anthropology and Forensic Entomology

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Lesson A: Determining Ancestry and Sex from Skeletal Remains

1. Identify four pieces of information that a forensic anthropologist would include in the osteobiography of a subject. (4 marks)

2. Describe three ways that human skeletal remains can be differentiated from animal skeletal remains. (3 marks)

3. Describe two ways that the sex of skeletal remains can be determined when no skull has been found. (2 marks)

Lesson B: Determining Traumatic Injuries from Skeletal Remains

4. Why is the cause of death usually inferred by forensic anthropologists rather than confirmed after they have studied the skeletal remains? (3 marks)

5. Complete the chart below by stating one type of weapon that could have caused each type of trauma, one type of force that produces each type of trauma, and the type or types of bone injury resulting from each type of trauma. (12 marks)

Type of Trauma	Type of Weapon	Type of Force	Type(s) of Bone Injury
<i>Blunt Force</i>			
<i>Projectile</i>			

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Sharp Force			
Strangulation			

Lesson C. The Use of Forensic Entomology in Criminal Investigations

6. How can insects be used to determine if an individual has suffered from poisoning? (2 marks)

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Use the following information to answer the next question.

The skeletal remains of an adult male were found under some tree branches and snow on the outskirts of Edmonton in January. A forensic entomologist observed more blowfly pupal membranes below the ribs in the chest region than in the skull of the victim.

7. a. Use your knowledge of insect maturation or development to explain that the victim described above was deposited in the above location before or after winter began. (2 marks).

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b. What does the observation of numerous pupal membranes in the chest region suggest about the cause of death of the victim described above? (2 marks)

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8. a. What type of database did the Canadian Police Research Centre begin in 1996? (1 mark)

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b. Why is such a database necessary in Canada? (2 marks)

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Lesson D. Crime Case Studies Involving Forensic Anthropology and Forensic Entomology

Case Study #1: Bones that were Worth a Thousand Words

10. a. Identify two pieces of skull evidence that would have distinguished Czar Nicholas' wife, Czaritsa Aleksandra, from the adult male skeletons? (2 marks)

b. Outline two pieces of evidence other than bone evidence used to identify the remains of Czaritsa Aleksandra. (2 marks)

Case Study #2: The Girl, the Army Sergeant, and the Bugs

11. a. Explain why no beetles were found on the victim's skeletal remains. (2 marks)

b. Why did forensic entomologists in this case obtain data from a weather station near the crime scene? (2 marks)

Case Study #3: The Farmhouse Murder

12. List three pieces of evidence that suggest the victim had been dead for longer than 20 days. (3 marks)